

The Critical Maneuverability Theorem

A First-Principles Design Framework for Agile Supercavitating Vehicles

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My Bio

- Henry Shen, Stevens Tech Class of 2029
- Pinnacle Scholar, Engineering w/ Naval Concentration, Pure and Applied Mathematics
- Research focus: machine learning architecture, group theory, random matrix theory and applications.
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Introduction: The Supercavitating Torpedo Paradigm

- **What is Supercavitation**

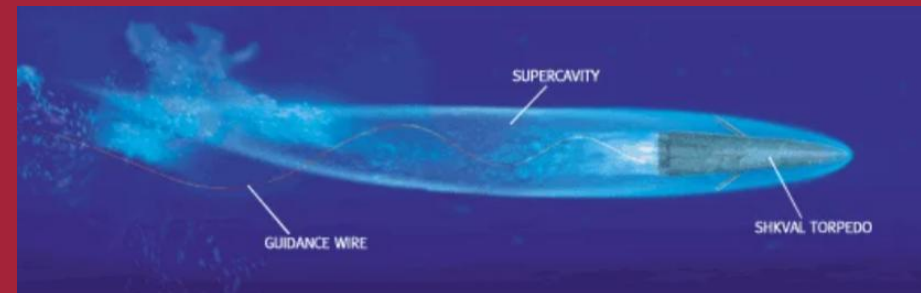
- A hydrodynamic technique that envelops a vehicle in a continuous gas bubble, reducing skin friction drag by up to 90%.
- Enables underwater speeds exceeding 200 knots (e.g., the VA-111 Shkval and Diehl BGT Barracuda).

- **The Vision for Naval Dominance**

- SCTs are a fundamentally different class of underwater weapon.
- Their extreme speed compresses adversary response time from minutes to seconds.

- *References:*

- *Polmar & Moore (2004) [1] — Cold War Submarines.*
- *Friedman (2006) [2] — The Naval Institute Guide to World Naval Weapon Systems.*



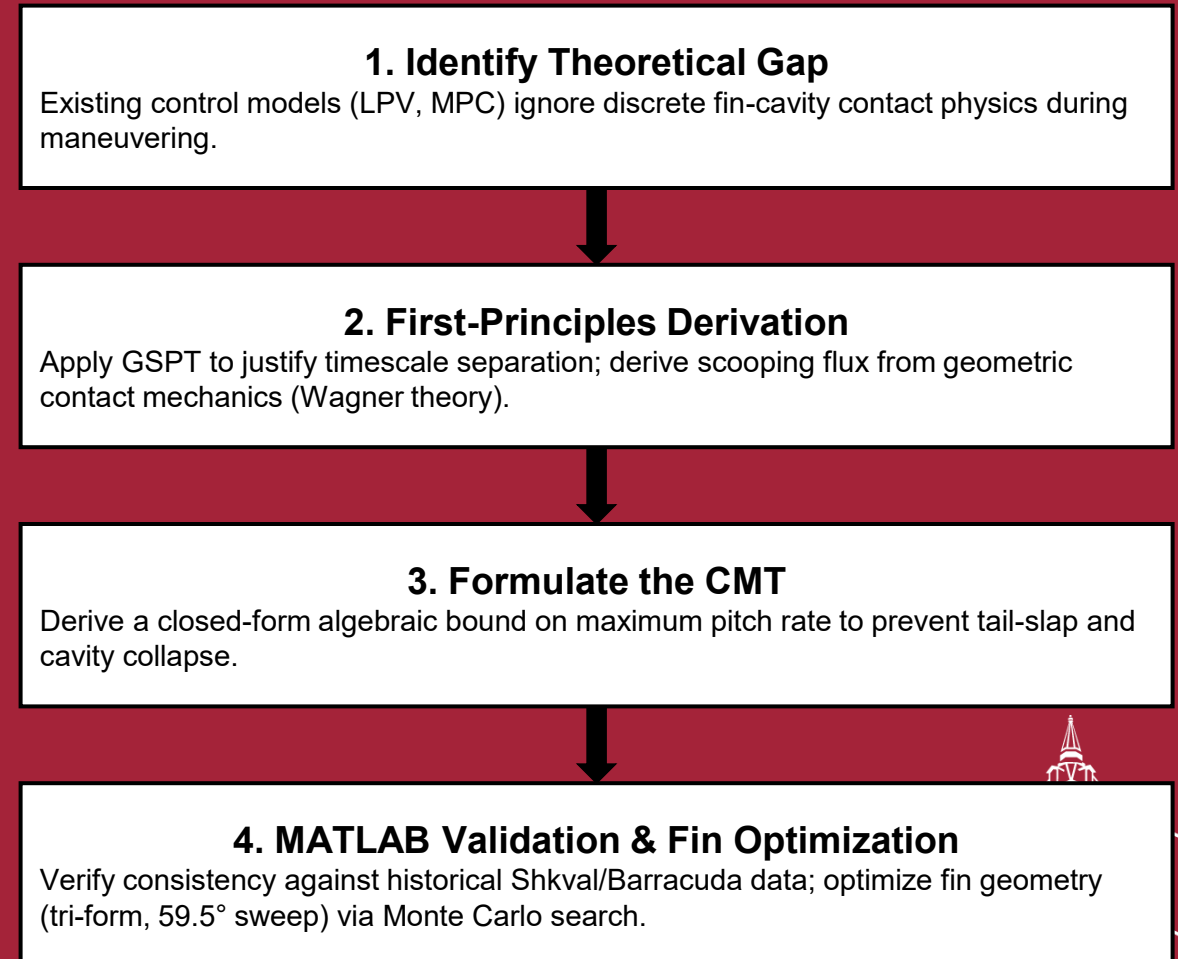
Motivation

- For SCTs, while steady-state cavity maintenance is well understood, maneuvering is highly unstable. Existing designs are practically straight-runners.
- A fundamental, unresolved trade-off exists between speed and maneuverability.
- SCTs are controlled by nose deflection, gas injection, and fins.
- Maneuvering requires control surfaces to penetrate the cavity wall.
- Traditional models ignore the discrete, multi-phase impact physics of fin penetration.
- Fin penetration acts as a mechanical scoop, ejecting gas from the cavity and triggering rapid, irreversible cavity collapse.



Research Goal: The CMT

- Derivation of a closed-form limit on maneuverability from first principles.
- Establish the Critical Maneuverability Theorem (CMT) to serve as a real-time constraint.
- Provide explicit design guidelines prioritizing vehicle and fin geometry to maximize agility.



Theoretical Framework

- Mass conservation of the cavity gas phase under isothermal conditions:
 - $d/dt(P_c V_{cav} / R_s T) = \dot{m}_{in} - \dot{m}_{out}$
- Non-dimensionalization reveals a singularly perturbed system (detailed on next slide).
- Geometric Singular Perturbation Theory (GSPT) shows that pressure dynamics equilibrate $O(10^{-3})$ faster than pitch dynamics.
- Mathematically justifies the quasi-steady gas assumption ($Q_{in}^* = Q_{out}^*$).

$$\frac{d}{dt} \left(\frac{P_c V_{cav}}{R_s T} \right) = \dot{m}_{in} - \dot{m}_{out}$$



Geometric Singular Perturbation

- **Step 1: Governing ODE (from Eq. 2)**

- $V_{cav}/(R_s \cdot T) \cdot dP_c/dt = \dot{m}_{in} - \dot{m}_{out}$

- **Step 2: Non-Dimensionalize**

- Define scales: $P_0 = P^\infty$, $V_0 = L^3$, $t_0 = L/V$, $\dot{m}_0 = \rho L \cdot V \cdot L^2$
- Substituting into the ODE, the leading coefficient becomes:
- $(V_0 \cdot P_0) / (R_s \cdot T \cdot \dot{m}_0 \cdot t_0) = P^\infty / (R_s \cdot T \cdot \rho L) = \rho g / (\rho L) = \varepsilon$

$$\frac{d}{dt} \left(\frac{P_c V_{cav}}{R_s T} \right) = \dot{m}_{in} - \dot{m}_{out}$$

- **Step 3: The Singularly Perturbed System**

- $\varepsilon \cdot dP'/d\tau = \hat{m}_{in}(P', \alpha) - \hat{m}_{out}(P') \quad (\varepsilon \approx 10^{-3})$
- $d\alpha/d\tau = g(\alpha, P') \quad (O(1) \text{ dynamics})$

- **Step 4: Tikhonov's Theorem ($\varepsilon \rightarrow 0$)**

- As $\varepsilon \rightarrow 0$, the fast variable P' instantaneously equilibrates:
- $0 = \hat{m}_{in} - \hat{m}_{out} \rightarrow Q_{in} = Q_{out}$ (slow manifold)
- This algebraic constraint replaces the PDE, enabling real-time computation.



Geometric Contact & Cavity Penetration

- With the quasi-steady assumption justified (Slide 7), the remaining task is to quantify forced gas loss when fins breach the cavity wall.
- Control surface (fin) penetration triggers contact when vehicle angle of attack exceeds a geometric threshold:
- $\alpha_{contact} = (R_{cav} - R_{tail}) / L$
- Penetration depth $\delta(\alpha)$ scales linearly with distance from the cavity nose:
- $\delta(\alpha) = L(\alpha - \alpha_{contact})$
- Mechanical contact leads to structural loads and cavity disturbance.

$$\alpha_{contact} = \frac{R_{cav} - R_{tail}}{L}$$



Scooping Flux & Curvature Correction

- Fin penetration drives high-speed liquid into the cavity interior (Wagner spray root):
- $\dot{m}_L = \rho_L \cdot V \cdot \Phi_{fin} \cdot L^2 \cdot (\alpha - \alpha_{contact})^2 \cdot \tan(\theta_{wedge})$
- **Note: ρ_L appears because this is the liquid mass influx rate, not the gas mass loss.**
- By momentum-driven entrainment ($\dot{m}_L \cdot V \approx \dot{m}_g \cdot V$), gas mass evacuated $\approx \dot{m}_L$.
- Converting to gas volumetric demand via $\div \rho_g$ yields the ρ_L/ρ_g factor in the CMT.
- Curvature correction for cylindrical cavity geometry:
- $\chi_{topo} = (\pi/2)^2 \cdot [1 - (4L\alpha)/(\pi R_{cav})]$
- Ref: Wagner (1932) [6]; Kornev & Mazur (2021) [7]; Korobkin (2004), J. Fluid Mech.

$$\dot{m}_{scoop} = \rho_L V \Phi_{fin} L^2 (\alpha - \alpha_{contact})^2 \tan \theta_{wedge}$$

$$\chi_{topo}(\alpha) = \left(\frac{\pi}{2}\right)^2 \left(1 - \frac{4}{\pi} \frac{L\alpha}{R_{cav}}\right)$$



Synthesis & Agility Envelope

- Cavity collapse occurs dynamically when gas demand exceeds ventilation valve capacity:
 - $C_{Q,req} = C_{Q,valve}$
- Solving this balance yields the Critical Maneuverability Theorem (CMT) limit on the maximum angle of attack:
 - $\alpha_{max} = \alpha_{contact} + \sqrt{(C_{Q,valve} - C_{Q,steady}) / (...)}$
- Establishes a Thermodynamic Hard Deck defining the boundary of the safe flight envelope.
- Design guideline: prioritizing vehicle and fin geometry is essential to maximize vehicle agility.

$$\alpha_{max} = \alpha_{contact} + \sqrt{\frac{C_{Q,valve} - C_{Q,steady}}{\frac{\rho L}{\rho_g} \Phi_{fin} \chi_{topo} \left(\frac{L}{D_n}\right)^2 \tan(\theta_{wedge})}}$$



Consistency Verification

- Validated against public data for two historical supercavitating vehicles:
 - VA-111 Shkval: CMT predicts collapse at $\alpha_{\text{crash}} = 5.16^\circ$, matching its historical straight-runner turn limit ($\sim 5^\circ$).
 - Diehl Barracuda: predicts collapse at $\alpha_{\text{crash}} = 4.79^\circ$.
- This thinner cavity validates the necessity of the Barracuda's active swiveling nose to extend the envelope.

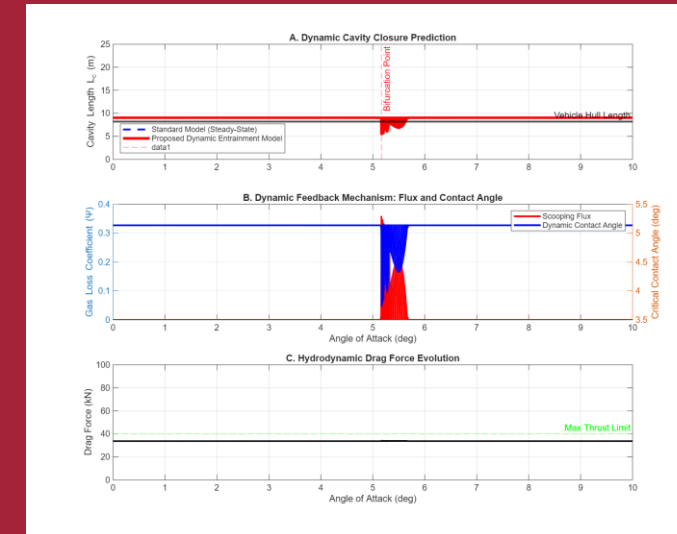


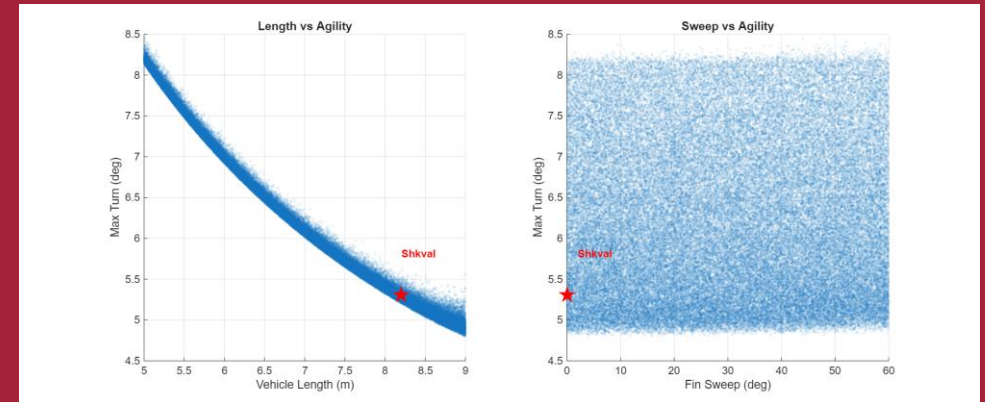
TABLE I
KEY SIMULATION PARAMETERS FOR HISTORICAL VEHICLE VALIDATION

Parameter	Shkval	Barracuda	Unit
Vehicle Length, L	8.20	7.00 (est.)	m
Hull Diameter, D_{hull}	0.533	0.53	m
Cavitator Diameter, D_n	0.10	0.08 (est.)	m
Cruise Velocity, V	100	111	m/s
Gas Supply Rate, $\dot{m}_{gas}$	1.20	1.30 (est.)	kg/s



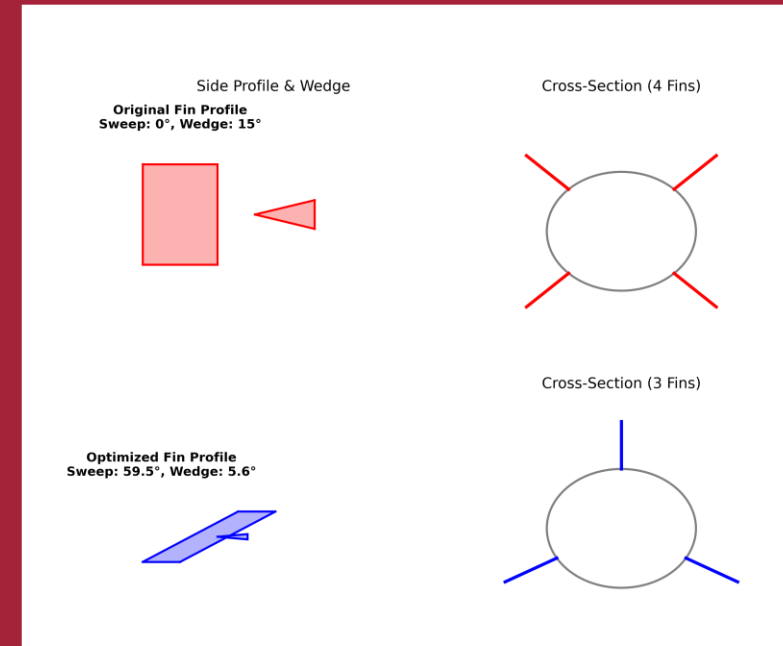
Sensitivity & The Length Penalty

- Global sensitivity analysis shows vehicle length L is the dominant parameter. This is geometrically straightforward—the non-obvious insight is that fin design is the primary actionable lever.
- Agility is quadratically penalized by length (L^2 in the denominator of the CMT).
- Shrinking vehicle length from 8.2 m to 6.0 m would achieve $\alpha_{\max} > 7.0^\circ$ under identical gas supply, proving that geometric scaling is the primary bottleneck.



Fin Geometry Optimization

- Monte Carlo search (100k iterations) conducted to maximize $\alpha_{\max}$ for a Shkval retrofit ($L = 8.2$ m).
- Optimal configuration: Tri-form tail (3 fins) with high sweep (59.5°) and sharp wedge (5.6°).
- Achieves $\alpha_{\max} = 5.71^\circ$ (7.5% improvement over cruciform).
- High sweep reduces effective solidity at initial impact; 3 fins balance CMT performance against 6-DoF control authority.



Insights

- **The Traditional View: Relying on Gas Supply**
- Historically, improving maneuverability meant increasing gas injection ($C_{Q, \text{valve}}$) to avoid cavity collapse.
- However, gas supply is physically bounded by internal volume and weight constraints, offering diminishing returns.
- **The CMT Paradigm: Geometry is the Key**
- The Critical Maneuverability Theorem reveals that scooping flux is linearly dependent on fin solidity (Φ_{fin}) and wedge angle.
- Optimizing fin geometry minimizes gas ejection at the source, effectively doubling the number of sequential turns before gas depletion.
- *Key Insight: Gas supply is physically bounded, but intelligent fin design increases turning capability without additional mass or volume penalties.*



Integration & Generalization

- CMT evaluates in $O(1e-4)$ seconds, allowing direct inclusion in 100 Hz complex control loops as a hard safety constraint.
- Generalizes to articulating noses:
 - $\alpha_{max} = (\alpha_{contact} + \delta_{nose}) * [1 + \sqrt{\dots}]$
- Explains how nose deflection ($\delta_{nose} \approx 12^\circ$ for Barracuda) dramatically extends the flight envelope.

$$\alpha_{max} = (\alpha_{contact} + \delta_{nose}) \left[1 + \sqrt{\frac{C_{Q, valve} - C_{Q, steady}}{\frac{\rho_L}{\rho_g} \Phi_{fin} \chi_{topo} \left(\frac{L}{D_n}\right)^2 \tan(\theta_{wedge})}} \right]$$



Limitations & Future Work

- **Current Limitations**

- Idealized Thermodynamics: The CMT currently assumes an isothermal cavity and uniform boundary curvature.
- Simplified Contact Mechanics: Fin-cavity contact is modeled as a macroscopic volumetric scooping flux, which does not resolve the 3D multiphase splashing at the fin-cavity interface.

- **Future Work**

- Experimental Validation: Conduct water-tunnel testing to empirically quantify the scooping flux coefficient during dynamic fin penetration across varying wedge/sweep angles.
- Advanced Control Integration: Embed the CMT algebraic bounds directly into a full nonlinear MPC architecture.
- 6-DoF Validation: Test the integrated CMT-MPC controller in a full 6-DoF underwater simulation environment to demonstrate real-time evasion of catastrophic tail-slap.



Thank you!

Questions?

